GEOSPATIAL ASSESSMENT | KHU-01 | 2024-04-01

Batiaghata Union Coastal Cropland

Baseline Report

Batiaghata, Khulna · Centroid 22.7301°N, 89.4921°E

Report ID	**BA-2025-20260817-E15B07**
Prepared by	**BDCrops AgroVision™**
Acquisition	**2024-04-01 · SENTINELHUB · cloud <5%**
Generated	**2026-08-17**
Status	**QA PASS | Production Tier | CONFIDENTIAL**

True-colour composite (Sentinel-2 L2A) — Batiaghata Union Coastal Cropland, Khulna — 2024-04-01

AOI	Cloud	Resolution	QA Status
KHU-01	<5%	10.0 m	PASS

Scene acquired **2024-04-01** via **SENTINELHUB**. over **Batiaghata Union Coastal Cropland** (Batiaghata, Khulna).

1. Report Specifications

AOI / Topic	Batiaghata Union Coastal Cropland	**Date**	2024-04-01
AOI Code	KHU-01	**AOI ID**	AOI_KH01
Centroid	22.7301°N, 89.4921°E	**AOI Bounds**	89.442147, 22.680075, 89.542147, 22.780075
EPSG	4326	**Source File**	AOI_Khulna_L2.kmz
Provider	SENTINELHUB (production)	**Cloud**	<5%
Source Resolution	10 m	**Bands**	—
Analysis Resolution	10.0 m	**QA Status**	■ PASS (Production Tier)
Report ID	BA-2025-20260817-E15B07		

1.1 AOI Footprint Summary

Bounding Box (WGS84)	minLon 89.442147, minLat 22.680075 maxLon 89.542147, maxLat 22.780075
Centroid (lat, lon)	22.7301°N, 89.4921°E
CRS / EPSG	EPSG:4326
Source	AOI_Khulna_L2.kmz

1.2 Executive Summary

• Scene **Sentinel-2 L2A** acquired **2024-04-01** via **SENTINELHUB**, cloud <5%, QA **PASS**.

• Dominant land cover: **Vegetation** 38.2%, **Agriculture** 37.8%, **Fallow** 10.4%.

• Detailed analytics for water, LULC, change detection, drainage, soil carbon, and crop residue follow on subsequent pages.

2. Compliance Matrix

*Verifies scene-level compliance with thresholds — resolution, cloud cover, projection, bit depth, band availability and orthorectification. **PASS** indicates all minimum requirements met.*

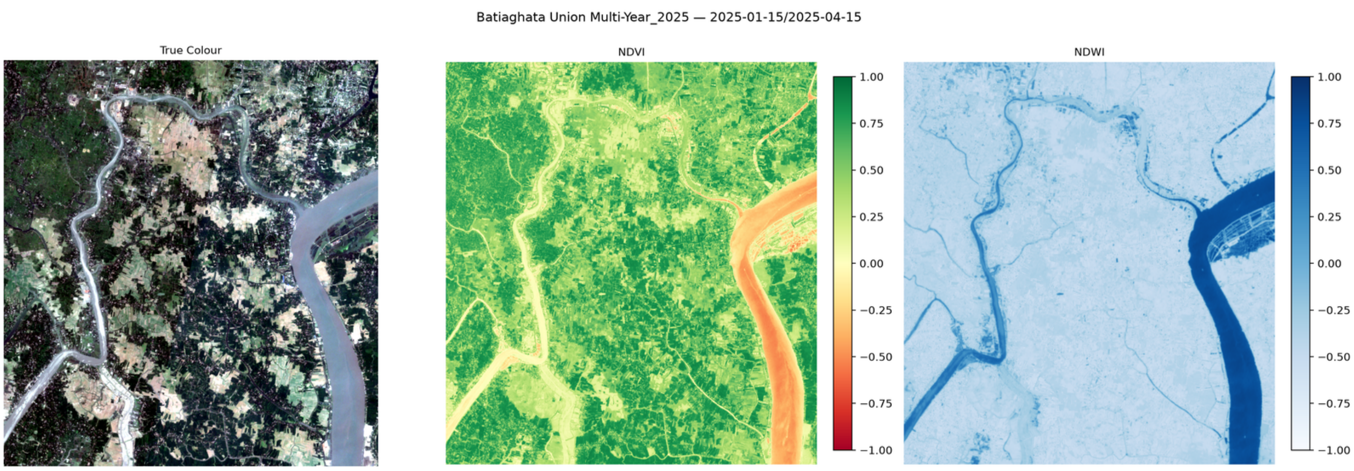
Tier	Status	Resolution	Cloud	Bands	Projection
FAIL	**■ PASS**	**10.0 m**	**<5%**	**0**	**EPSG:4326**

#	Check	Threshold	Actual	Status
1	Cloud Cover	≤ 5 %	<5%	PASS
2	Resolution (spectral preferred)	≤ 10 m	10.0 m	PASS
3	Resolution (multispectral preferred)	≤ 1.5 m	10.0 m	FAIL
4	Bands acquired	4 required (Blue, Green, Red, NIR)	0 acquired	FAIL
5	Projection	EPSG:4326 / UTM	EPSG:4326	PASS
6	Bit Depth	16-bit	16-bit	PASS
7	Orthorectified	Required	Yes	PASS
8	QA Overall	PASS	PASS	PASS

3. Spectral Indices

Per-pixel ratios of satellite bands that highlight specific land conditions: vegetation health (NDVI / EVI / SAVI / NDRE), surface water (NDWI / MNDWI), and tillage / residue proxies. Each index is summarised over the valid (non-cloud) pixel area.

NDVI	NDWI	NDBI	EVI	SAVI	NBR
0.458	**-0.426**	**-0.164**	**0.331**	**0.292**	**0.347**



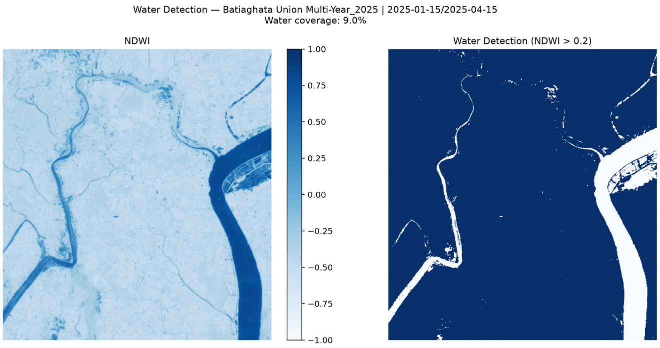
Spectral indices dashboard — NDVI, NDWI, EVI, SAVI

Index	Mean	Std Dev	Min	Max
NDVI	0.4582	0.3030	-0.6103	0.8982
NDWI	-0.4262	0.2749	-0.8242	0.6818
NDBI	-0.1645	0.1879	-0.7748	0.6459
EVI	0.3314	0.2165	-0.2069	1.0000
SAVI	0.2920	0.1839	-0.1818	0.7110
NBR	0.3471	0.2095	-0.5472	0.8408
MNDWI	-0.3050	0.3176	-0.6825	0.8421
NBR2	0.2047	0.0815	-0.0991	0.3958

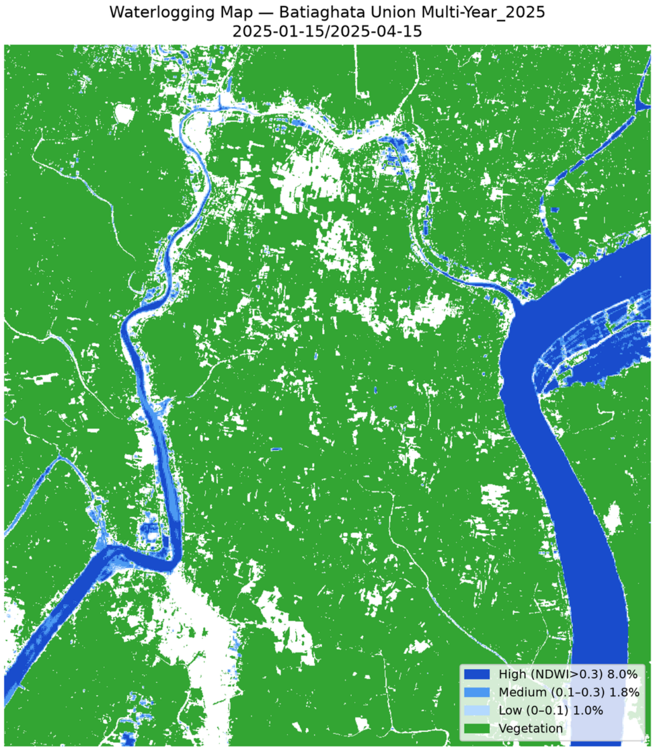
4. Water & Waterlogging Analysis

*Detects open water and inundated cropland using NDWI (Green / NIR) and MNDWI (Green / SWIR) spectral indices. **Permanent Water** = pixels consistently water-covered. **Flood Risk** = low-lying areas with elevated water signatures relative to dry-season baseline.*

Permanent Water	Flood Risk	Enhanced Water	MNDWI Used
4.24%	**4.13%**	**8.06%**	**Yes**



Water Detection Map

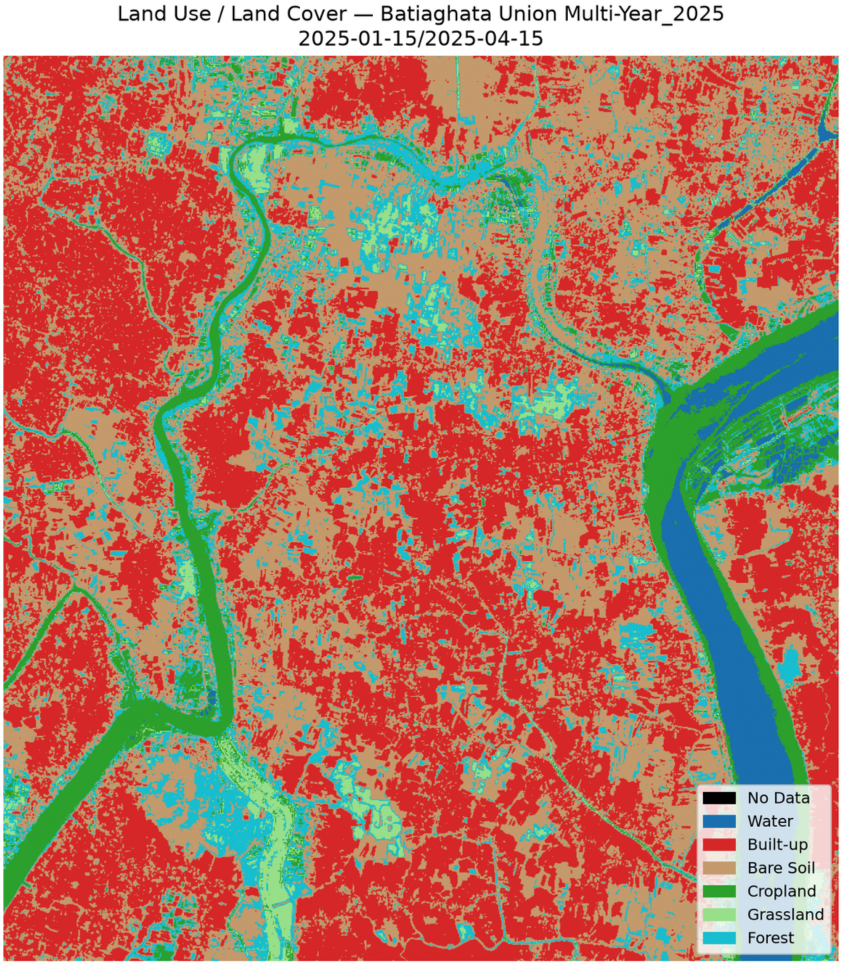


Waterlogging Risk Map

Metric	Value
Permanent Water (NDWI > 0.3)	4.24%
Flood Risk Zone	4.13%
Enhanced Water (MNDWI / fused)	8.06%
MNDWI Index Available	Yes

5. Land Use / Land Cover (LULC)

Pixel-level classification of the AOI into thematic classes (water, vegetation, agriculture, bare soil, fallow). Built from spectral signatures using the available band stack; SWIR-required Built-up class merges into Bare Soil when SWIR is unavailable.



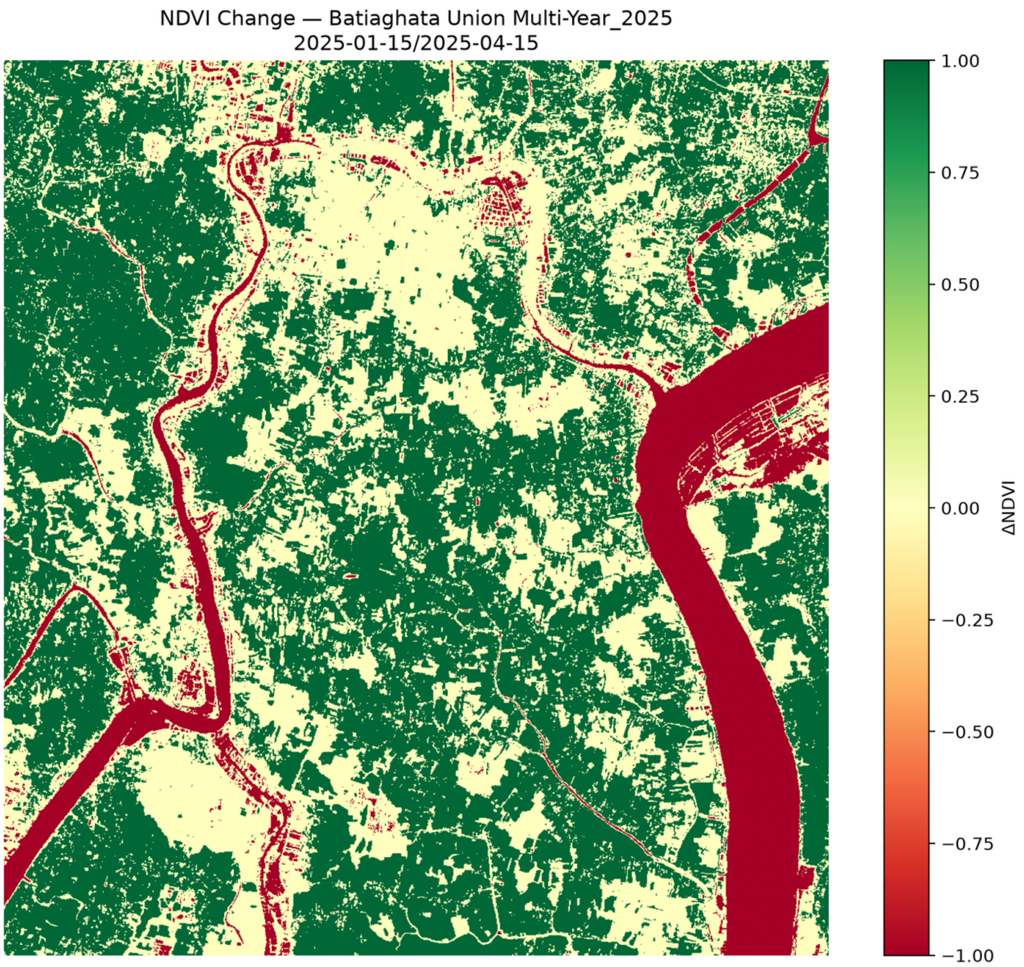
LULC Classification Map

Swatch	Class	Area %	Hex
	Water	4.24%	#1F77B4
	Vegetation	38.19%	#2CA02C
	Agriculture	37.82%	#FFEB3B
	BareSoil	6.74%	#8C564B
	BuiltUp	2.61%	#D62728
	Fallow	10.39%	#9467BD

6. Change Detection

Compares NDVI between two acquisition dates to quantify vegetation gain, loss and stability. Synthetic-baseline mode flags single-date runs where no real prior-date scene was supplied — results in that case are indicative only.

Mode	Gain	Loss	Stable	Mean Δ
Synthetic (single date)	**51.23%**	**10.99%**	**37.79%**	**0.1582**



NDVI Change Map

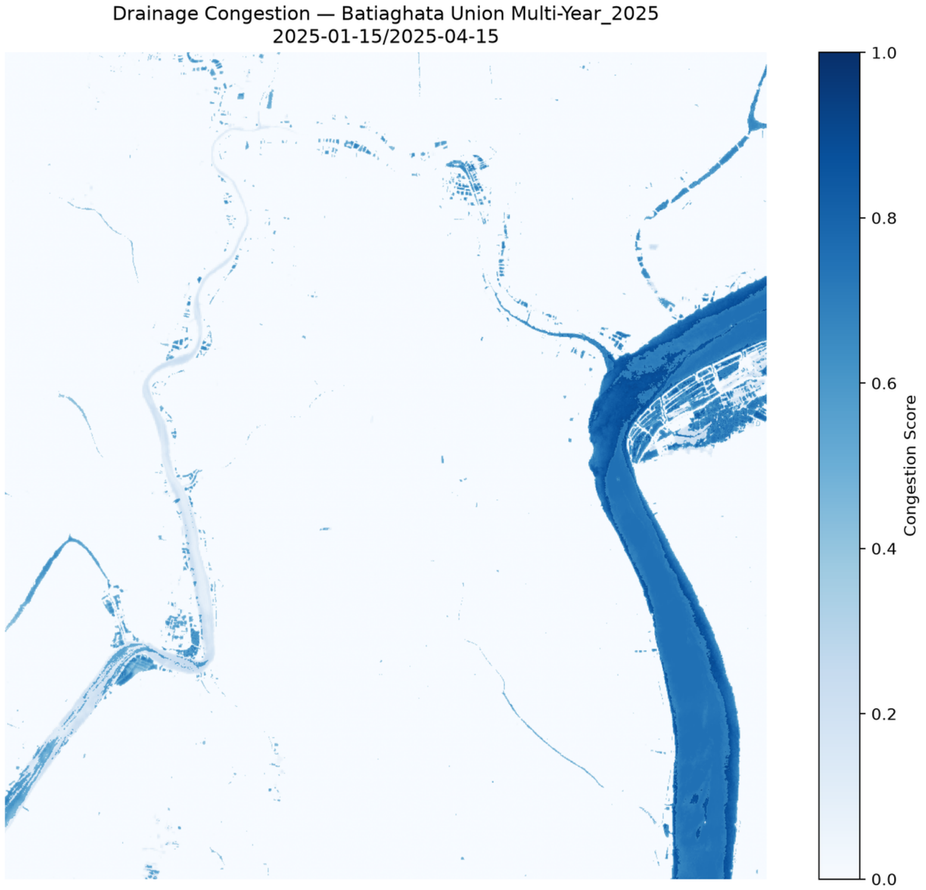
Mode	Gain %	Loss %	Stable %	Mean Δ	Gain Thresh	Loss Thresh	Synthetic Baseline
Synthetic (single date)	51.23%	10.99%	37.79%	0.1582	0.200	0.200	0.3

■ *Synthetic baseline used — no dual-date imagery provided; results are indicative only.*

7. Drainage Analysis

Identifies areas of persistent surface-water accumulation that indicate poor drainage. Congestion score combines NDWI persistence with terrain proxies; full Topographic Wetness Index requires a DEM.

Congestion %	Persistence	Threshold	TWI Used
7.38%	**0.060**	**—**	**No**



Drainage congestion map — blues scale, persistence proxy (no DEM)

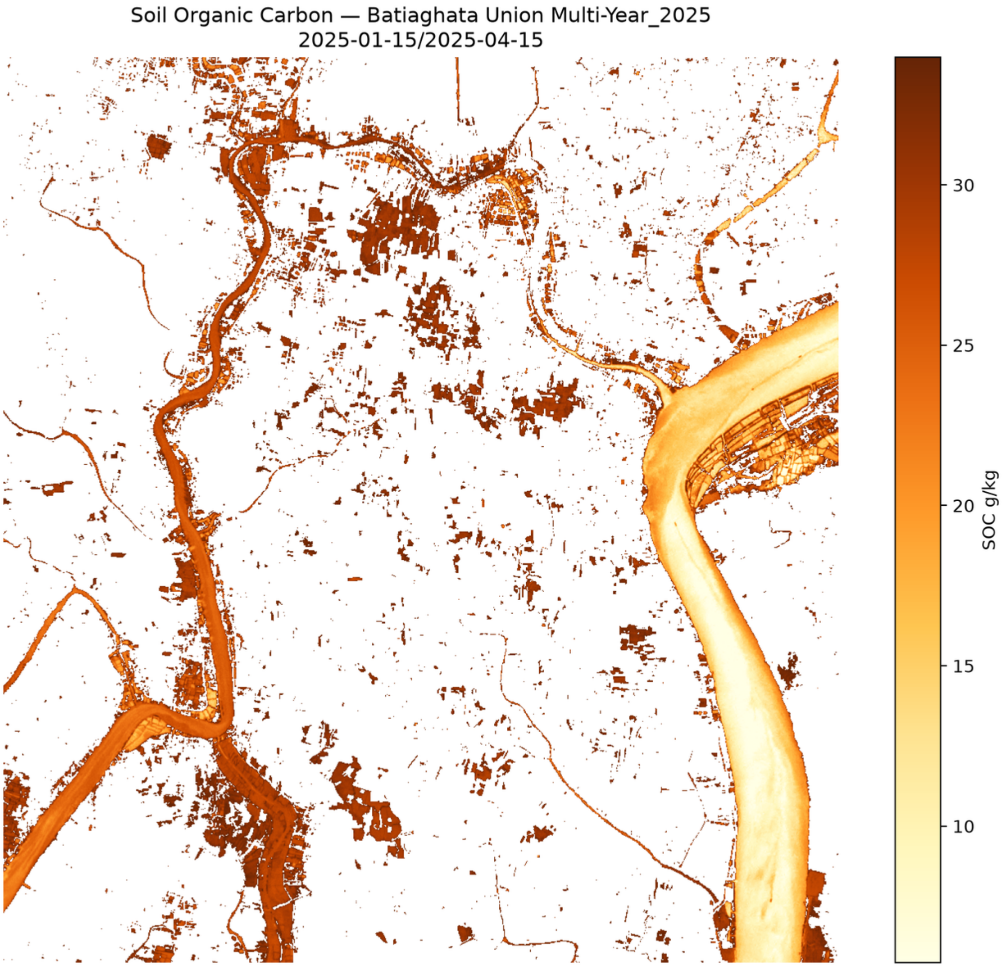
Zone	Class	Congestion %	Persistence	Pixels
LULC	Water	99.55%	0.718	48428
LULC	Vegetation	0.00%	0.000	436123
LULC	Agriculture	0.00%	0.000	431900
LULC	BareSoil	45.92%	0.413	76976
LULC	BuiltUp	2.26%	0.053	29810
LULC	Fallow	5.00%	0.005	118701

■ *No DEM supplied — TWI deferred; congestion score uses persistence-only weighting.*

8. Soil Organic Carbon

*Estimates Soil Organic Carbon content (g/kg) over bare-soil pixels via regional spectral calibration. Mode **regional** uses a fixed national / regional model; **field** mode would use ground-truth samples (not provided in this run).*

SOC Mean	SOC Std	Uncertainty	Bare-soil	Mode
24.03 g/kg	**8.85 g/kg**	**—**	**229169**	**regional**



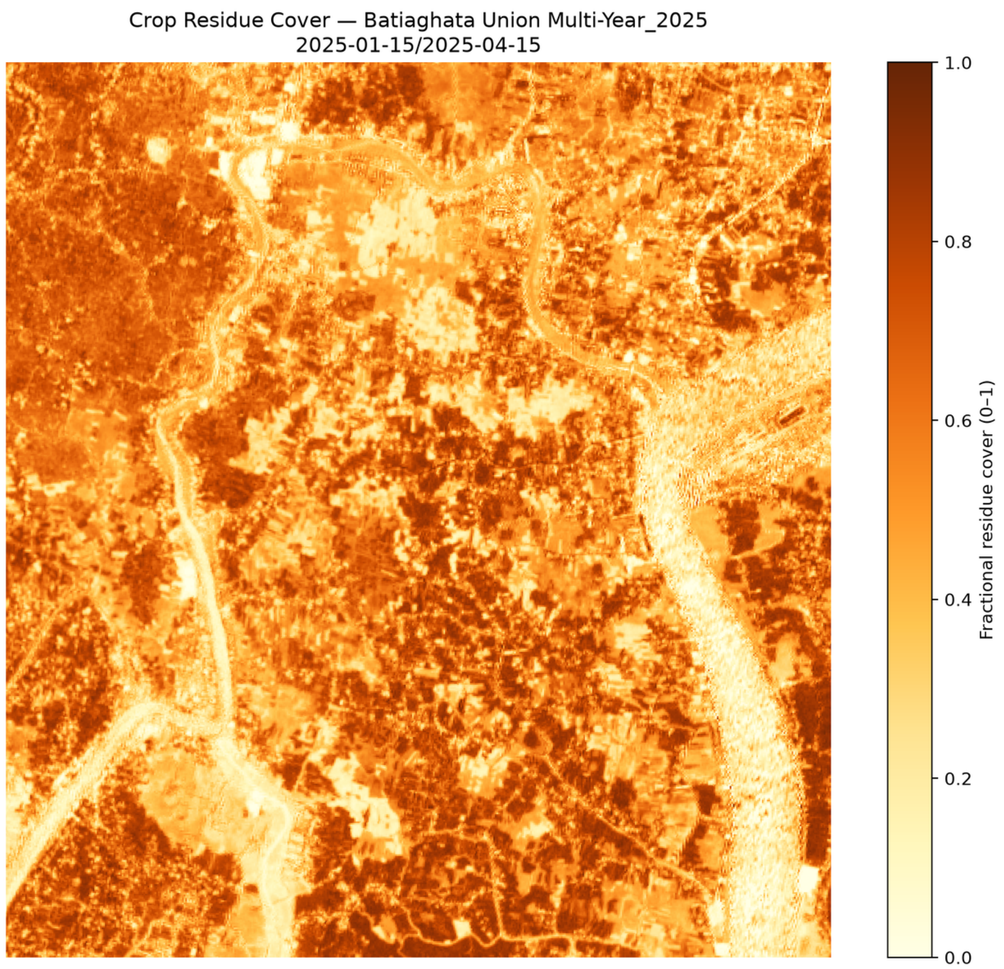
Soil Organic Carbon Density Map

Zone	Class	SOC Mean (g/kg)	SOC Std (g/kg)	Calib Mode	R^2
ALL_BARE	bare_soil	24.033	8.849	regional	—
LULC	Water	9.945	3.058	regional	—
LULC	BareSoil	23.253	5.009	regional	—
LULC	BuiltUp	28.308	2.294	regional	—
LULC	Fallow	32.348	1.085	regional	—

9. Crop Residue

Estimates fractional residue cover over agricultural pixels. NIR/Red ratio is used as a proxy when SWIR bands are unavailable; high-accuracy NDTI / STI requires SWIR (Sentinel-2 B11 / B12 or Landsat-9).

Index Used	Mean Cover	High Cover >30%	Valid Pixels
NDTI	**51.20%**	**83.39%**	**1,141,938**



Crop Residue Cover Map

Zone	Class	Mean Residue Cover	High Cover %	Valid Pixels	Index Used
ALL	all_valid	0.5120	83.39%	1,141,938	NDTI
LULC	Water	0.1920	18.10%	48,428	NDTI
LULC	Vegetation	0.7053	99.96%	436,123	NDTI
LULC	Agriculture	0.4714	95.14%	431,900	NDTI
LULC	BareSoil	0.2826	43.36%	76,976	NDTI
LULC	BuiltUp	0.1840	14.58%	29,810	NDTI

Zone	Class	Mean Residue Cover	High Cover %	Valid Pixels	Index Used
LULC	Fallow	0.3113	49.63%	118,701	NDTI

■ *Residue estimated via NIR/Red ratio — SWIR bands unavailable; interpret with caution.*

10. Key Findings

Auto-generated narrative summarising the most significant metrics, exceedances and data caveats from the preceding analytics. The full machine-readable trace is preserved in `geodata_run_result.json`.

GeoData Findings — Batiaghata Union Multi-Year_2025

Generated by Unit-GeoData

---

1. Acquisition Summary

- **Status:** `success` ■
- **Provider:** `sentinelhub`
- **Satellite:** —
- **Scene ID:** –
- **Acquisition window:** —
- **Cloud cover:** —
- **Resolution:** 10.0 m
- **Bit depth (delivered):** —-bit
- **Projection:** — / —
- **Bands acquired:** 6/6 ['B02', 'B03', 'B04', 'B08', 'B11', 'B12']

2. Quality Assurance

Check	Result
`bands_ok`	■
`bit_depth_ok`	■
`cloud_ok`	■
`files_ok`	■
`ortho_ok`	■
`projection_ok`	■
`resolution_ok`	■
`resolution_preferred_ok`	■
PASS	■

Warnings (1):

- ■■ cloud_cover_pct not reported by provider

4. Spectral Indices (Stage 3)

- **Status:** `success` ■
- **Provider:** sentinelhub
- **Computed:** 8 indices
- **Skipped:** 1 (provider missing required bands)

4a. Generic Indices

Index	Min	Max	Mean	Range	Note
`NDVI`	-0.610	0.898	0.458	■	Vegetation health
`NDWI`	-0.824	0.682	-0.426	■	Surface water / vegetation moisture
`NDBI`	-0.775	0.646	-0.164	■	Built-up area (needs SWIR)
`EVI`	-0.207	1.000	0.331	■	Enhanced vegetation index (atmosph ere-corrected)
`SAVI`	-0.182	0.711	0.292	■	Soil-adjusted vegetation (sparse cover)
`NBR`	-0.547	0.841	0.347	■	Normalized burn ratio
`MNDWI`	-0.683	0.842	-0.305	■	Modified NDWI (water discrimination)
`NBR2`	-0.099	0.396	0.205	■	

4b. WV-3-specific Indices

(not computed — provider is `sentinelhub`*. NDRE/CRI/NMDI/NBAI require WorldView-3 8-band imagery.)*

5. Compliance Matrix

Requirement	Spec	Actual	Status	Ref
Resolution	≤ 0.5 m	10.0 m	■	§4.1

Requirement	Spec	Actual	Status	Ref
Cloud cover	≤ 5.0%	—	■■	§4.2
Projection	UTM / WGS84	—	■	§4.3
Bit depth (final)	16-bit	None-bit	■	§4.4
Bands present	0 ()	6	■	§4.5
Orthorectified	required	yes	■	§4.3
Overall	passed_qa	PASS	■	—

6. §5 Deliverables — Gap Analysis

Section	Deliverable	Tool	Status	Tender Ref	Next Steps
§5.1	Soil Organic Carbon	`soil_carbon`	■ disabled in profile	§5.1	Enable in `geodata_tools` to surface gap
§5.1	Crop Residue Cover	`crop_residue`	■ disabled in profile	§5.1	Enable in `geodata_tools` to surface gap
§5.2	Drainage Congestion	`drainage`	■ disabled in profile	§5.2	Enable in `geodata_tools` to surface gap

7. Generated Visualizations

PNG outputs (11):

- `png/change_map.png`
- `png/dashboard.png`
- `png/drainage_map.png`
- `png/false_color.png`
- `png/lulc_map.png`
- `png/qa_overlay.png`
- `png/residue_map.png`
- `png/soc_map.png`
- `png/true_color.png`
- `png/water_detection.png`
- `png/waterlogging_map.png`

Report generated by CF2 Unit-GeoData · 2026-08-17T23:07:33 Full machine-readable trace: `geodata_run_result.json` *(14,521 bytes)*

11. Glossary of Terms

Reference definitions for every acronym and technical term used across this report. Terms appear in the order they are first encountered above; entries are alphabetical within this glossary. Region: Bangladesh.

Term	Definition
8FYP	Eighth Five-Year Plan (2020-2025). GoB development framework with dedicated Agriculture, Water and Climate chapters.
ADB	Asian Development Bank. Multilateral lender co-financing climate, irrigation and coastal-embankment projects.
AEZ	Agro-Ecological Zone. Land classification dividing Bangladesh into 30 zones based on physiography, soils, hydrology and climate.
AEZ-13	Ganges Tidal Floodplain. Coastal saline zone covering greater Khulna, Satkhira, Bagerhat. Rice-shrimp mosaic dominates.
AIS	Agriculture Information Service. MoA public communications wing producing farmer advisories, print and broadcast content.
Aman	Bangladesh monsoon rice season. Transplanted Jul-Aug, harvested Nov-Dec. Rain-fed, largest cropped area.
AOI	Area of Interest. The geographic region delimited by the bounding box for which all analytics in this report are computed.
Aus	Bangladesh pre-monsoon rice season. Direct-seeded or transplanted Mar-Apr, harvested Jun-Aug. Rain-fed, drought-prone.
BADC	Bangladesh Agricultural Development Corporation. State agency for seed multiplication, irrigation infrastructure and input distribution.
BARI	Bangladesh Agricultural Research Institute. National institute for non-rice crops — wheat, maize, pulses, oilseeds, vegetables.
Barind	Elevated Pleistocene terrace in NW Bangladesh. Drought-prone; managed by BMDA.
BBS	Bangladesh Bureau of Statistics. Official statistical agency; publishes crop area, production estimates and agricultural census.
BCCSAP	Bangladesh Climate Change Strategy and Action Plan. National framework for food security, agricultural adaptation and DRR.
Beel	Perennial or seasonal freshwater lake in low-lying floodplain. Important for fisheries and dry-season boro paddy.
BINA	Bangladesh Institute of Nuclear Agriculture. Develops crop varieties via induced mutation and biotechnology.
Bit depth	Number of bits per pixel per band in the source imagery. Higher bit depth (e.g. 16-bit) preserves finer reflectance gradations.
BJRI	Bangladesh Jute Research Institute. National centre for jute, kenaf and allied fibre research.

Term	Definition
BMDA	Barind Multipurpose Development Authority. Development agency for the Barind Tract; groundwater irrigation and afforestation.
Boro	Bangladesh dry-season rice. Transplanted Dec-Feb, harvested Apr-Jun. Irrigated, highest yielding season.
BRAC	Largest NGO in Bangladesh. Extensive rural livelihood, microfinance and agriculture programmes reaching millions of smallholders.
BRRI	Bangladesh Rice Research Institute. Develops BRRI dhan varieties (e.g. dhan28, dhan29, dhan47, dhan52, dhan89, dhan100).
BRRI dhan100	Zinc-enriched Boro rice variety (2021). Biofortified with ~25 mg/kg zinc addressing micronutrient deficiency.
BRRI dhan29	High-yielding Boro rice variety. 160-day duration, yield potential 7.0-7.5 t/ha under optimal irrigation.
BRRI dhan47	Salinity-tolerant Boro rice variety (2007). Tolerates soil salinity up to 12-14 dS/m; suited to coastal saline zones.
BRRI dhan52	Submergence-tolerant Aman rice variety carrying the SUB1 gene. Survives 14+ days of complete submergence.
BRRI dhan89	Modern Boro rice variety (2018). Yield potential 8.0-8.5 t/ha, 158-day duration, superior grain quality.
BSRI	Bangladesh Sugarcane Research Institute. Research and variety development for sugarcane.
BWDB	Bangladesh Water Development Board. Manages coastal polders, embankments, drainage sluices and river-training works.
CEGIS	Center for Environmental and Geographic Information Services. Semi-autonomous body providing GIS/RS services on water, land, agriculture.
Char	River island or emerging sand-silt land formed by riverine deposition. Highly productive but flood-prone.
CIMMYT	International Maize and Wheat Improvement Center. CGIAR centre supporting Bangladesh wheat, maize and conservation-agriculture research.
CIP-BD	Country Investment Plan of Bangladesh. Government-endorsed 5-year framework aligning agriculture, food security and nutrition investments.
Cloud cover %	Estimated fraction of the AOI obscured by cloud at acquisition time, as reported by the provider or computed from the scene quality mask.
Congestion score	Drainage-section metric combining NDWI persistence with terrain proxies. Higher values indicate areas prone to standing water.
DAE	Department of Agricultural Extension. MoA field-level extension arm delivering advisory services via Upazila and Union offices.
EPSG	EPSG Geodetic Parameter Dataset code identifying the coordinate reference system. EPSG:4326 = WGS-84 geographic (lat / lon).

Term	Definition
EVI	Enhanced Vegetation Index. Atmosphere-corrected variant of NDVI; less prone to saturation in dense canopy. Range -1 to +1.
FAO-BD	Food and Agriculture Organization of the UN, Bangladesh Country Office. Technical support to MoA on statistics, crop monitoring and food security.
Flood Risk	Pixels with elevated water signatures relative to dry-baseline; indicates seasonal inundation or low-lying areas vulnerable to flooding.
GCF	Green Climate Fund. UNFCCC financing mechanism; funds Bangladesh coastal adaptation and climate-resilient agriculture.
Haor	Bowl-shaped tectonic depression in NE Bangladesh (Sylhet, Sunamganj). Deeply flooded during monsoon; single Boro crop annually.
HYV	High-Yielding Variety. Modern improved cultivars requiring irrigation and fertiliser inputs.
iDE	International Development Enterprises. Market-based NGO active in Bangladesh irrigation, vegetables and rural entrepreneurship.
IFAD	International Fund for Agricultural Development. UN agency financing smallholder agriculture; major char and coastal portfolio in Bangladesh.
IRRI	International Rice Research Institute. CGIAR centre with long-standing Bangladesh programme on rice varieties, stress-tolerance and mechanisation.
L2A	Sentinel-2 Level-2A product — bottom-of-atmosphere surface reflectance, atmospherically and topographically corrected.
LGED	Local Government Engineering Department. Rural infrastructure — farm-to-market roads, small-scale irrigation, drainage.
LULC	Land Use / Land Cover. Pixel-level classification into thematic classes (water, vegetation, agriculture, bare soil, fallow, etc.).
MNDWI	Modified NDWI. Uses SWIR instead of NIR; better separation of water from built-up surfaces. Requires SWIR band (B11 / B12).
MoA	Ministry of Agriculture, Government of Bangladesh. Umbrella ministry over DAE, BADC, BARI, BRRI, SRDI, and related institutions.
MoDMR	Ministry of Disaster Management and Relief. Coordinates flood, cyclone and drought response; overlaps with agricultural risk management.
NAEP	National Agricultural Extension Policy. GoB framework for DAE service delivery, farmer-field-schools, and PPP extension linkages.
NAP-BD	National Adaptation Plan of Bangladesh (2023-2050). UNFCCC-registered plan with agriculture-sector adaptation priorities.
NDRE	Normalized Difference Red-Edge. Uses the Red-Edge band (B05); sensitive to canopy chlorophyll content — useful for crop-stress detection.
NDTI	Normalized Difference Tillage Index. Detects crop residue using SWIR1 / SWIR2 bands. High accuracy; SWIR-required.

Term	Definition
NDVI	Normalized Difference Vegetation Index. Ratio of NIR and Red bands; sensitive to live green vegetation. Range -1 to +1.
NDWI	Normalized Difference Water Index. Detects open water using Green and NIR bands. Range -1 to +1 (positive = water).
NIR / Red ratio	Near-Infrared / Red reflectance ratio. Fallback proxy for crop residue when SWIR bands are unavailable. Indicative accuracy only.
Permanent Water	Pixels consistently water-covered across the analysis window — distinct from seasonal flood-risk zones.
PKSF	Palli Karma-Sahayak Foundation. Bangladesh government wholesaling microfinance to NGO-MFIs for rural livelihoods.
Polder	Embanked coastal land parcel managed by BWDB. Bangladesh has 139 polders totalling ~1.2M ha reclaimed from tidal floodplain.
Preview Tier	Lower-resolution / lower-cost imagery tier used for preliminary scoping. Field-level decisions typically require Tasked tier.
QA	Quality Assurance. Per-scene compliance checks (cloud cover, bit depth, band availability, projection, orthorectification).
Resolution	Ground sample distance (pixel size) of the source imagery. Sentinel-2 native: 10 m (visible/NIR), 20 m (SWIR/RE), 60 m (atmospheric).
SAVI	Soil-Adjusted Vegetation Index. NDVI variant correcting for soil-brightness; useful when vegetation cover is sparse (<40%).
Sentinel-2	ESA optical Earth-observation mission providing 13-band multispectral imagery at 10-60 m resolution with 5-day global revisit.
SOC	Soil Organic Carbon (g/kg). Carbon stored in soil organic matter; indicator of soil health, fertility and water-retention capacity.
SPARRSO	Space Research and Remote Sensing Organization. Bangladesh national remote-sensing agency; agricultural mapping and disaster monitoring.
SRDI	Soil Resources Development Institute. Bangladesh authority for soil surveys, salinity monitoring and AEZ mapping.
STI	Soil Tillage Index. SWIR-based residue cover proxy; complements NDTI for tillage / residue mapping.
Synthetic baseline	Change-detection mode used when only a single date of imagery is available — a baseline is statistically synthesised. Indicative only.
TWI	Topographic Wetness Index. DEM-derived index identifying low-lying areas with high water-accumulation potential. Required for high-accuracy drainage congestion mapping.
WARPO	Water Resources Planning Organization. Under Ministry of Water Resources; national macro-level water planning authority.

12. Recommendations & Next Steps

Auto-derived action items addressing each gap or exceedance found in earlier sections: procurement upgrades, supplementary data acquisition (DEM / SWIR), and follow-up analytical work.

1. **Procurement:** Activate Maxar/WorldView tasking to meet 1.24 m spec. Sentinel-2 fallback is QA-compliant but limits field-level mapping.

2. **Baseline:** Synthetic baseline used — schedule a dual-date acquisition (e.g. pre/post-monsoon) for verifiable change-detection metrics.

3. **QA Metadata:** Provider did not report cloud_cover_pct. Switch to a provider that exposes per-scene cloud statistics for full QA traceability.